



Fayoum STEM school

Sustainable Design and Implementation of a Savonius VAWT for Renewable Energy Storage

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Chapter One

Egypt's grand challenges

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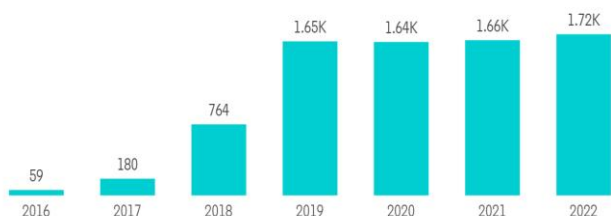
Chapter 1

Egypt's grand challenges

Improving the use of alternative energy.

Egypt, a country with a lot of natural resources has high potential in development of the alternative energy section. Egypt with a perfect location, vast deserts, high solar irradiance, and consistent wind patterns is a home for alternative energy production projects to be implemented. However with all these advantages, Egypt energy sector strongly relies on energy production from fossil fuels, which account for the majority of energy production. Dependence like this not only contributes to environmental degradation but also exposes the country to volatility of the global oil and gas markets. To overcome these challenges Egypt must set improving the use of alternative energy as a priority to work on and develop, benefiting from the solar, wind, and hydro-power energies.

Egypt Solar Energy Market : Solar Energy Installed Capacity, in MW, Egypt, 2016-2022



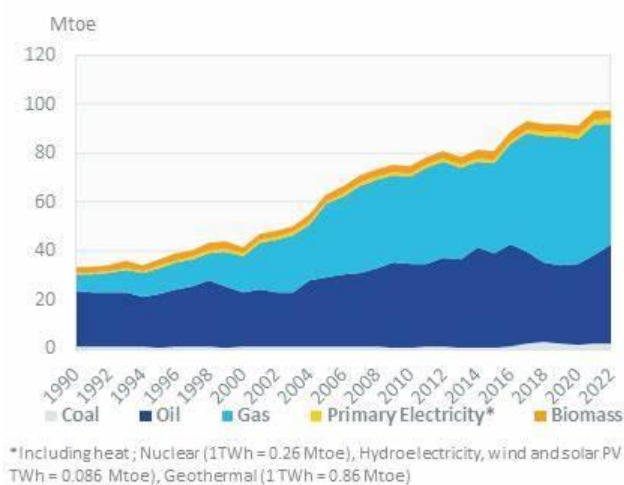
Source: International Renewable Energy Agency

Graph 1

Egypt's energy landscape is undergoing a transformation, driven by the need to meet growing energy demand, reduce greenhouse

gas emissions, and enhance energy security. Egypt has made progress in this field in recent years, particularly in the development of solar and wind energy projects. The Benban Solar Park, located in the Aswan Governorate, is one of the largest solar installations in the world, with a total capacity of 1.8 gigawatts (GW). This project, along with several wind farms in the Gulf of Suez, has significantly increased Egypt's renewable energy capacity. Hydro-power also plays an important role in Egypt's energy production with the high dam in Aswan as an example. The dam generates approximately about 2.1 Gigawatts of electricity contributing the base of the country's supply of power. Egypt started also exploring other energy sources such as: Biomass, and Geothermal energy but these still in the early stages of development.

On the contrary, renewable energy still accounts for a small portion in Egypt's total energy production. According to the International Renewable Energy Agency (IRENA), renewable energy sources contributed approximately 9% of Egypt's electricity generation in 2020. This is expected to increase in the future as the government implement its ambitious renewable energy targets to be 42% of the total energy production by 2035.



Graph 2

- **Obstacles to the adoption of the alternative energy:**

- 1) Technical challenges:**

One of the primary obstacles is interference of renewable energy sources, such as: solar, and wind. Unlike fossil fuels which offers consistent and reliable energy supply, solar and wind energy depends on atmospheric conditions. This volatility can cause problems in energy supply. With that the infrastructure system is primarily designed for power generation from fossil fuels, making it difficult to integrate between renewable and non-renewable energy systems.

- 2) Financial challenges :**

The high primarily costs of connected to renewable energy projects, while the long term operational costs of renewable energy systems are lower than that of the fossil based energy systems. Moreover, access to financing for renewable energy projects is often limited.

- 3) Social and cultural factors :**

Public awareness and acceptance is crucial for renewable energy technologies adoption and implementing, In Egypt, there's a lack in the awareness of the alternative energy production importance, and the severity of the energy production from fossil fuels, specifically in rural areas. Furthermore, the transition from fossil fuel to renewable energy may face obstacles from the fossil fuel industry.

- **Strategies for improving the use of alternative energy :**

To overcome the obstacles and accelerate the adoption of alternative energy sources, Egypt must implement a comprehensive strategy that addresses technical, financial and social challenges, and here are some methods of

improving the use of alternative energy in Egypt:

- 1) Enhancing the infrastructure :**

The infrastructure is crucial in the improvement of the alternative energy, as this step will allow more alternative energy to flow through the total production of energy in Egypt.

- 2) Increasing access to investment :**

This will allow the flow of money through the alternative energy projects, which will make the production of alternative energy rise and refresh the market.

- 3) Strengthening the Regulation:**

An effective and transparent regulatory environment is key to attracting investment to the renewable energy sector. The Egyptian government can simplify the approval process for renewable energy projects, minimizing obstacles, and have clear and consistent policies. The government can also consider implementing more ambitious renewable energy goals and incentives.

- 4) Promoting public awareness and education :**

Increasing public awareness is critical for the adoption of the alternative energy. The Egyptian government should fund campaigns and initiatives in schools and universities to educate people raise their awareness and remove the misconceptions about renewable energy and its reliability.

- 5) Encouraging International Cooperation:**

Egypt should utilize its geographical location and wealth of renewable energy resources to invite international investment and

cooperation. The country can join regional and international renewable energy initiatives, like the Desertec Industrial Initiative and the International Solar Alliance, to share knowledge, technology, and best practices. In addition, Egypt should pursue the formation of partnerships with top renewable energy companies and research institutions to advance the development and implementation of new renewable energy technologies.

- **Previous efforts of The Egyptian government:**

Egypt has known for many years that it needs to change to alternative energy sources. Egypt would like to use more renewable energy since it needs more energy, wants to be less reliant on fossil fuels, and needs to save the environment. Egypt has made numerous plans through the years, started huge projects, and worked with other countries to help create and use alternative energy

- 1) **Zafarana Wind Farm:**

The Zafarana Wind Farm, located in the Gulf of Suez, is a project in Egypt's renewable energy portfolio. With a total capacity of 545 MW, Zafarana is one of the largest wind farms in Africa. The project was developed in multiple phases, with the first phase commissioned in 2001. Zafarana has played a crucial role in



Figure 1

demonstrating the viability of wind energy in Egypt and has contributed to the country's efforts to reduce greenhouse gas emissions.

- 2) **Aswan High Dam and Hydropower:**

Aswan High Dam is an important part of Egypt's renewable energy projects, though not a new one. It was constructed in 1970 and can produce 2.1 gigawatts of hydropower. It supplies much electricity for Egypt's basic needs. In recent years, the government has been looking for ways to refurbish and increase the dam's production of hydropower to keep on serving the country's energy needs.



Figure 2

- 3) **European Union (EU):**

The EU has been of great assistance to Egypt's renewable energy efforts. They have assisted with finances and technical support on various projects. One example is the contribution made by the EU towards the Gabal El Zeit Wind Farm construction. This wind farm has a capacity of producing 580 MW of power and is one of the biggest wind farms in Africa.

- 4) **Feed-in Tariff Program (2014):**

In 2014, Egypt introduced a Feed-in Tariff (FiT) program to incentivize private sector investment in renewable energy. The program offered guaranteed prices for electricity generated from solar and wind projects,

providing investors with a stable and predictable revenue stream. The FiT program was initially launched in two phases, with the first phase targeting 2,300 MW of solar and wind capacity.

While the FiT program faced some challenges, including delays in project implementation and financing issues, it succeeded in attracting significant interest from both domestic and international investors. The program also helped to raise awareness about the potential of renewable energy in Egypt and laid the groundwork for future initiatives.

Improve the scientific and technological environment for all.

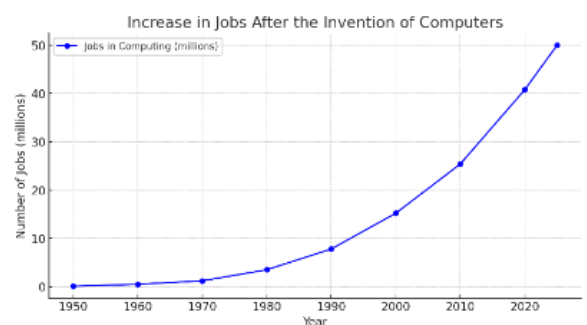
Improving the scientific and technological environment for all people in our community is a very important thing we should do and care about it. As it has a lot benefits, such as:

1. **Increasing the access to opportunities:** increasing the scientific and technological environment in our community make people gain skills to build their career in the emerging fields like AI, programming and etc.
2. **Accelerated Innovation and Problem Solving:**
 1. **Global collaboration:** improving the scientific and technological environment for all can lead to collaboration between people from different cultures to solve global problems like global warming, climate change, food security and etc.

2. **Creating differences in points of view:** improving the scientific and technological environment leads to making different opinion which makes solving problems being in hand of people who has different opinions not in hand of specific people who have the same opinion. This can help creating better solutions for the problems that we face.

3. Economic Growth and Development:

- **Job creation:** improving the scientific and technological environment leads to creating new inventions that need new jobs. For example, creating computers created new jobs like programming and computer engineers as you see in this figure. As we know these jobs were created since in the last century.



Graph 3

- **Improving industries:** As we know, if we devolved the technology we can improve the industry which make us create more money and improve our economy like what happen in the industrial revolution.

4. Improved Quality of Life:

improving the scientific and technological environment can lead to improving the quality of life in many cases such as:

- **Health care:** with improving the technology of medicines, we can have a better treatments and treat the diseases that we don't know how to cure yet like cancer. Which help us save lots of lives
- **Enhanced living standards:** advanced technology like smart houses, cleaner sources of energy , robots and etc can make our life easier and safer.

5. Social and Cultural Advancement:

- **Culture exchange:** technology can help people communicate at any place and any time which make people from different cultures communicate and collaborate to solve global problems.

To achieve the benefits of improving the scientific and technological environment for all here are some key strategies we can follow:

1. Enhance the access of education and training:

- **Expand STEM Education:** Make science, technology, engineering, and mathematics (STEM) education more accessible, starting from early education through to higher learning. This can include introducing hands-on science activities in schools, online courses, and scholarships for underrepresented groups.
- **Offer Skill Development Programs:** Encourage continuous learning through workshops, online platforms, and community-based training to equip people with the skills required to engage with new technologies.

2. Promote Inclusivity in Science and Technology:

- **Encourage Diversity in STEM:** we need to work on decreasing gender , racial disparities in the stem field like offering scholarships and creating welcoming environments for them so they can

develop themselves and be able to improve our community.

- **Support Marginalized Groups:** Provide resources and support to help marginalized communities access technology and scientific knowledge, including specialized training programs and outreach initiatives.

3. Improve Access to Technology and Infrastructure:

- **Making it easier to having internet access:** if we made it easier to internet access this can help people to learn more and develop themselves.
- **Making technology less expensive:** if we made technology easier, all people would be able to have technology that can help them develop themselves like computers and cell phones.

4. Encourage Collaboration and Open Science:

- **Making initiatives:** governments should make global initiatives that help people increasing their skills with technology and science and these initiatives should be available for all people from all over the world.
- **Improving self-learning:** Government should make self-learning more easier. They can accomplish that by making open sources websites where people can learn from it for free. They also should encourage programmers to make more websites like these. For example NASA website. Nasa website give the ability for people to learn more about space.

And now, after we knew about the benefit and how to accomplish the improving of the scientific and technological environment for all, lets know the bad effect of not improving the scientific and technological environment.

If we didn't achieve this goal, we would face a negative consequences. Here are some of these consequences:

1. Widening inequality:

- The community gap; if all people are not equal in access to education and technology, a big gap will appear between them. As a result, they won't be equal in the opportunity and this will lead to leaving a lot of people behind.
- Global dividing: people in low-income areas will be away from getting the benefit of the technology and knowledge which leads to leaving them without access to the important services like health care and education. Or in other meaning they will live like they are in hell.

2. Slower Progress and Innovation:

- **Stagnation:** without improving the education and technology, this will make a terrible problems. As we will miss the developments of important services like health care and not solving the global problems like global warming.
- **Missed opportunities:** not improving the education and technology will lead to not improving the industries that can make our life easier like AI, quantum computing and solving global problems. Can you imagine how our life would be if we stopped improving these industries?

3. Reducing the quality of life:

- **Limited access to modern services:** as no technology and scientific developments, services like health care and transportation will be less effective and accessible for the public
- **Increased Poverty and Unemployment:** as industry involves

with technology, people who don't have good education will not be able to work. as a result, more people will not be able to find jobs and this will affect the economy in a bad way.

And now, let me tell you the how this problem is related to our capstone.

Our capstone this semester is talking about generating energy from renewable resources to help our environment. So, to apply this in real life we need people to have a good scientific background and know how to use the modern technology. If people don't have neither good education nor modern technology we will miss the chance to improve this idea which can help us live an easier life and help our planet.

Address & reduce pollution in air, water and soil.

Egypt has 11 grand challenges, but our capstone projects focus on four of them mainly, one of these challenges that I will talk about now is Reducing air, water, and soil pollution, so I'll discuss the key sources of air, water, and soil pollution, their impacts, prior solutions, and provides some statistics to give you a clearer picture of the situation.

First of all, Air pollution is a significant environmental and public health issue in Egypt, particularly in urban areas like Cairo, Alexandria, and other industrial zones and it threatens Egypt, and it has many reasons that cause it like:

1- Vehicle Emissions

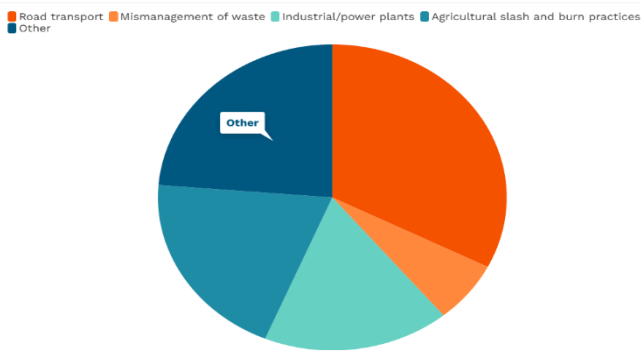
2- Industrial Emissions

3- Burning of Agricultural Waste

4- Dust and Sandstorms

5- Household Pollution

and there is a pie chart for the reasons of Air pollution:



Source: World Bank, Egypt cost of environmental degradation - Air and water pollution, 2019. • Notes: (i) The World Bank report aggregates 'Agricultural slash and burn practices' and 'Mismanagement of waste' into an 'Open Burning' category - the disaggregation was estimated based on data on 'Mismanagement of waste' from another source (McDuffie et al., PM2.5 Exposure Estimates, Sectoral Source Contributions, Total Attributable Mortality Estimates, and Fractional Disease Contributions, 2021); (ii) Secondary particulates in the primary source are assumed as PM2.5 emissions from 'Industry/power plants'.

Chart 1

Air pollution has a huge and harmful impact on Egypt in many fields like: Health, environment, economy, society, and culture.

1-Health Impacts:

Air pollution is one of the leading causes of preventable deaths and illnesses in Egypt, particularly in densely populated cities like Cairo and Alexandria.

2- Environmental Degradation:

Air pollution contributes to Nile River Pollution, Soil Contamination, Biodiversity Loss, and Climate Change Contribution.

3- Economic Costs:

Air pollution impacts the economy through Healthcare Burden, Lost Productivity, Tourism Decline, and Investment Deterrence.

4- Social and Cultural Impact:

Air pollution damages the Cultural Heritage like the damage that happens due to the acidic rains and particulate abrasion in the sphinx and the Karnak temple, Restoration costs for monuments exceed \$100 million annually.

Prior solutions:

1. Vehicle Emission Control

- Converted public vehicles to natural gas (50% by 2012) and introduced electric buses.
- Enforced stricter emission standards and fuel pricing reforms, reducing car sales by 35%.

2. Public Transit Expansion

- Invested \$17 billion in metro systems (Lines 1–3) and suburban railways to cut traffic congestion.

3. Agricultural Waste Management

- Recycled 90% of rice straw (500,000 tons in 2020) to prevent seasonal burning and "black clouds."
- Imposed fines up to \$6,300 and used satellites to monitor illegal burning.

4. Waste Management

- Rehabilitated landfills (e.g., Abou-Zaabal) and incentivized private-sector recycling.

Secondly, you know that water is very important for our life, and there is some defects happens to water in the last century caused water pollution by a huge amount.

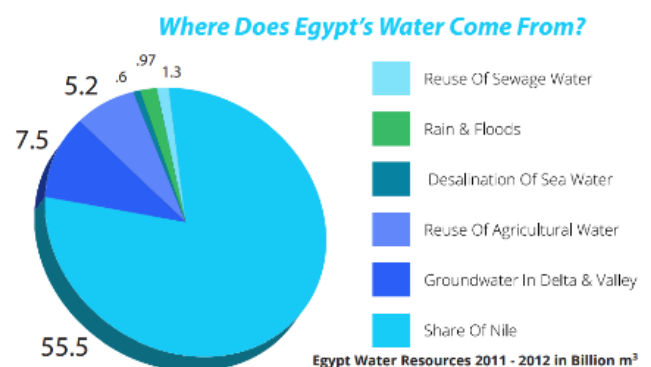


Chart 2

water pollution is a challenge that faces Egypt like air pollution, this problem is not less important than air pollution, if it does not have more importance, this problem has many reasons the most important of them are:

1- Agricultural Runoff (Primary Cause)

- Fertilizers, pesticides, and herbicides used in farming wash into rivers, lakes, and groundwater.

2- Industrial Discharges

- Factories and industries release harmful chemicals, heavy metals, and toxic waste into water bodies.

3- Oil Spills

- Spills from oil tankers, pipelines, or offshore drilling release large amounts of oil into oceans and rivers.

4- Plastic and Solid Waste

- Improper disposal of plastic waste, garbage, and other solid materials pollutes water bodies.

5- Acid Rain

- Air pollutants like sulfur dioxide and nitrogen oxides dissolve in rainwater, making it acidic.

Water pollution impacts on Egypt:

Water pollution in Egypt is causing serious health issues, damaging agriculture, and hurting the economy. Contaminated water spreads diseases, lowers crop yields, and impacts industries like fishing and tourism. It also harms ecosystems and worsens water scarcity. Poorer communities are hit hardest, with limited access to clean water. While solutions like better wastewater treatment exist, population growth and infrastructure challenges make it tough to solve. In short, pollution is a major threat to Egypt's health, economy, and environment.

Egyptian prior solutions:

Egypt is tackling water pollution through several key actions:

1. **Upgrading Wastewater Treatment:** The government is expanding and improving treatment plants to clean sewage before

it's released into water sources, and some treated water is being reused for farming.

2. **Stronger Pollution Laws:** There are stricter regulations for industries and agriculture to prevent harmful waste from polluting water, along with penalties for those who don't follow the rules.

3. **Raising Awareness:** Public education campaigns encourage people to conserve water, practice good hygiene, and dispose of waste properly to reduce pollution at the local level.

4. **Better Infrastructure:** Egypt is improving its water and drainage systems, especially in rural areas, to ensure clean water access and prevent untreated sewage from entering rivers and lakes.

5. **International Support:** Egypt is working with organizations like the World Bank to fund and implement better water management practices.

6. **Desalination:** To reduce pressure on the Nile, Egypt is investing in desalination plants to create fresh water from seawater, particularly for coastal areas. While these efforts are helping, Egypt still faces challenges in fully solving the water pollution problem due to population growth and limited resources.

Thirdly, Soil pollution in Egypt is a growing concern, as it directly impacts agriculture, public health, and the environment.

Reasons for Soil Pollution in Egypt:

1-Agricultural Chemicals

2- Industrial Waste

3- Sewage and Waste Disposal

4- Plastic Waste

Soil pollution impacts:

1- Reduced Agricultural Productivity: Soil pollution harms soil fertility, making it difficult for crops to grow. Contaminants like heavy metals can also accumulate in the food chain, leading to lower crop yields and poorer-quality food.

2- Health Risks: Polluted soil can lead to the contamination of food products, which can then affect human health. Long-term exposure to heavy metals and pesticides can lead to diseases such as cancer, kidney damage, neurological disorders, and developmental issues in children.

3- Loss of Biodiversity: Soil pollution damages ecosystems, making it harder for plants and animals to survive. This affects not only agricultural biodiversity but also the natural environment.

4- Decreased Water Quality: Pollutants in the soil can leach into groundwater, contaminating drinking water supplies and irrigation water. This further complicates water quality problems in Egypt.

Prior solutions:

1- sustainable Farming: The government is encouraging organic farming and sustainable practices, like crop rotation, composting, and integrated pest management (IPM). These methods reduce reliance on chemicals and help keep the soil healthy for future generations.

2- Soil Conservation Programs: Efforts are being made to prevent soil erosion, overgrazing, and deforestation. These programs aim to protect the soil and prevent it from degrading further, keeping land productive and healthy.

3- Educational Campaigns: Farmers and the public are receiving more training on sustainable farming practices. This includes

educating them about the dangers of excessive chemical use and the importance of protecting soil health for long-term agricultural success.

4- Government Initiatives for Crop Rotation: Farmers are being encouraged to practice crop rotation, a technique where different crops are planted in the same area over time to maintain soil fertility and avoid overuse of the same nutrients, which helps prevent soil degradation. *Graph 4*

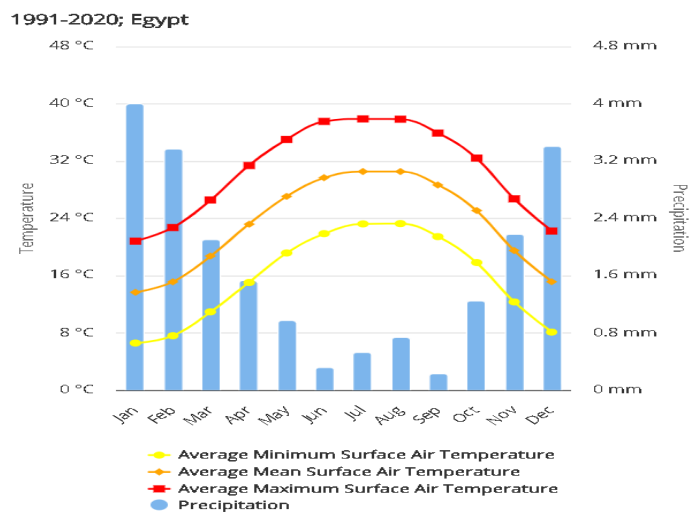
Reduce and adapt to the effect of the climatic change

Introduction:

Climate change refers to the long terms of temperature and weather patterns. Climate change consider one of the most important problems that doesn't face Egypt only, but it faces the whole world. So, it is considered a global challenge. With widespread consequences for the ecosystem, economies, and societies. It is primary driven by human activities since the industrial revolution. These

activities, such as burning fossil fuels, deforestation, and industrial processes. Which leads to increasing greenhouse gas (GHG) emissions such as Carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O). Which leads to an increase in the global temperature, melting glaciers in the two poles, more frequent and severe weather events, shifting ecosystems, and rising sea levels. These changes don't affect the natural environment only, they also affect human health, food security, and global economic development. In the subject of global warming, expansion in changed patterns of precipitation. (Graph 4) shows the Average minimum and maximum

rate of air temperature, and the precipitation rate in Egypt from 1991 to 2020.



Graph 4

Addressing this crisis requires urgent and comprehensive action on two fronts: mitigation and adaptation. Mitigation focuses on reducing the causes of climate change, such as cutting (GHG) emissions, and promoting renewable energy, while adaptation involves adjusting societies and ecosystems to cope with the changes that are already occurring. Tackling climate change is a shared responsibility that calls for international cooperation, innovative solutions, and grassroots action to ensure a sustainable future for both current and future generations.

Causes of the climate change in Egypt:

The causes of climate change can be broadly categorized into human-induced (anthropogenic) and natural causes. However, the current rapid changes in the Earth's climate are primarily driven by human activities, which



Figure 3

have intensified since the Industrial Revolution. Below are the key contributions:

Firstly, **Human-induced causes:**

1. Burning of Fossil Fuels:

- Fossil fuels like coal, oil, and natural gas are burned for electricity, transportation, and industrial processes.
- This releases large amounts of carbon dioxide (CO₂), the most significant greenhouse gas, into the atmosphere as in (Fig. 2)

Like: vehicles, and factories are major sources of CO₂ emissions.

2. Deforestation and Land Use Changes:

- Forests are natural carbon sinks that absorb carbon dioxide (CO₂) from the atmosphere and produce oxygen (O₂). (Fig. 4) represents the cutting down of the forests.



Figure 4

- Large-scale deforestation for agriculture, urban development, and logging reduces this capacity, while the burning of trees during land clearing releases stored carbon.

Like: Cattle farming and wet rice cultivation are significant methane contributors.

3. Agriculture and Livestock Farming:

- Methane (CH₄), a potent greenhouse gas, is emitted from livestock digestion (enteric fermentation) and rice paddies.

- The overuse of chemical fertilizers also leads to nitrous oxide (N₂O) emissions, another powerful greenhouse gas.

Like: Cattle farming and wet rice cultivation are significant methane contributors.

Secondly, **Natural Causes:**

1. Volcanic Eruptions:

- Volcanic eruptions release greenhouse gases like CO₂ and sulfur dioxide (SO₂). (Fig. 5) represents a volcanic eruption.



Figure 5

- Although their impact is small compared to human-induced emissions, they can contribute to short-term climate effects.

Like: the eruption of Mount Pinatubo in 1991 led to a temporary cooling effect due to the release of sulfur aerosols.

2. Solar Activity:

- *Variations in solar radiation due to sunspot cycles can influence global temperatures.*
- *However, the contribution of solar activity to current climate change is minimal compared to human activities.*

3. Natural Carbon Cycle Fluctuations:

- Oceans and forests naturally absorb and release CO₂, but disruptions (e.g., ocean warming) can limit their ability to act as carbon sinks.

The effects of climate change in Egypt:

1. Water scarcity:

- **Nile River Dependence:** Egypt largely depends on the Nile River for fresh water, which provides it with about 90% of its water. Climate change will reduce the flow of the Nile due to increased evaporation and changed rainfall patterns in the upstream countries.
- **Reduced Rainfall:** Egypt presently receives very little rainfall, and climate change will exacerbate droughts, thereby reducing water supplies even further. Which leads to Desertification as in (Fig. 5)



Figure 6

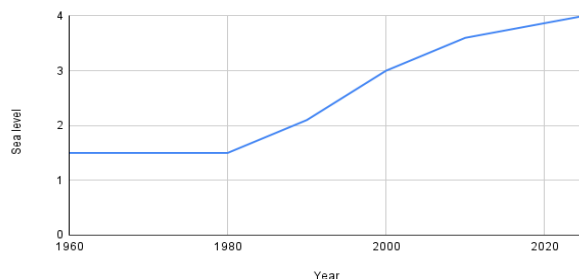
2. Agricultural Problems:

- **Crop Yields:** Rising temperatures and water deficits threaten agricultural production, particularly of staple crops like wheat and maize. This can lead to food insecurity and rising food prices.
- **Soil Salinization:** Sea level rise has the potential to cause saltwater intrusion into the Nile Delta, contaminating freshwater and lowering soil fertility.

3. Sea Level Rise:

- **Coastal Flooding and Erosion:** Egypt's northern coastline, including the Nile Delta, is already highly vulnerable to sea level rise. This can potentially displace millions, ruin infrastructure, and inundate rich agricultural land.
- **Habitat Loss:** Wetlands and mangroves are in danger of loss along the coasts, and this is at risk of jeopardizing biodiversity and fisheries. (Graph.5) represents the sea level rise in Egypt.

sea level rise in Egypt



Graph 5

4. Economic Impacts:

- **Tourism:** Egypt's economy, which is largely fuelled by tourism, could be impacted by the deterioration of beaches, coral reefs, and historical sites.
- **Fisheries:** A warmer sea and acidification could harm aquatic species, affecting the living of those who subsist on fishing.

5. Energy Demand:

- **Higher Cooling Needs:** Higher temperatures will correspondingly increase air conditioning demand, further straining Egypt's energy system and potentially increasing greenhouse gas emissions.

Reduce urban congestion

This grand challenge affects the whole world although all developments in world, many countries have urban congestion such as China, India and Egypt. Some factors that cause urban congestion are rapid population growth, unplanned urban expansion and lack of alternative cities and job opportunities.

Urban congestion causes many problems such as

- **Increasing transportation costs**
- **Increasing pollution**
- **Deterioration in health and psychological life**

There are many solutions for this problem such as

- 1) **Expanding and improving public transportation**
 - a) Metro expansion
 - b) Upgrading railway systems
- 2) **Encouraging decentralization and new cities development**
 - a) Developing new urban centers
 - b) Improving infrastructure in rural areas
- 3) **Building more bridges and tunnels**
- 4) **Mixed-use development**



Figure 7

Dealing with the population growth

Introduction:

Egypt's population has been growing up rapidly, reaching more than 105 million people, making it one of the most populous countries in Middle East and Africa. Egypt considered the 3rd in Africa and 14th globally. This rapid growth creates significant challenges, putting pressure on resources like water, food, housing, education, and healthcare. So, it also strains infrastructure and job markets. Managing population growth is a major challenge for Egypt as it works toward sustainable development and improving living standards.

Causes of population growth in Egypt:

1. *High Birth Rates:*

- Egypt continues to experience high birth rate compared to many other countries, particularly in rural areas where larger families are often preferred for cultural economic reasons.

2. *Improved Healthcare:*

- Advances in healthcare have led to lower mortality rates and longer life expectancy, contributing to a growing number of populations.

3. *Lack of Awareness and family planning:*

- Limited access to family planning and lack of awareness about the benefits of smaller families have contributed to population growth, especially in underserved areas and communities.

4. *Economic and Social Factors:*

- In some communities, children are viewed as a source of future financial security, encouraging families to have more children.

Effects of the population growth on Egypt:

Egypt's rapid population growth has had widespread effects on the country's resources, infrastructure, and economy. As the population continues to grow, these challenges are becoming more served. There are the effects of population growth in Egypt:

1. *Pressure on water resources:*

- Egypt is highly dependent on the Nile River water, but with the population growth increasing, the demand for water has exceeded sustainable levels.

2. *Food Insecurity:*

- The growing population has led t higher food demand, putting pressure on Egypt's agriculture sector, which is already limited by scarce land and water shortages.

3. *Urban Overcrowding and Housing Shortages:*

- Cities like Cairo and Alexandria are experiencing significant overcrowding, with increasing demand for housing leading to informal settlements and unsafe living conditions.

The prior solutions of the population growth in Egypt:

Egypt has implemented several solutions over the years address the challenges of rapid population growth. These efforts focused on raising awareness and improving access to family planning services. There are some prior solutions of the population growth in Egypt:

1. Family planning programs:

- The government introduced family planning initiatives to promote smaller families, particularly in rural areas.

2. “Two Is Enough” campaign:

- This national campaign, launched in 2018, encouraged families to limit the number of children to two.
- It aimed to change cultural norms around large families, especially in rural and less educated communities.

Improve sources of clean water

Improving the sources of clean water in Egypt is a critical issue, given the country's arid climate, limited freshwater resources, and growing population. Egypt relies heavily on the Nile River for its water supply, which accounts for

Egypt's water resources availability in billion cubic meter

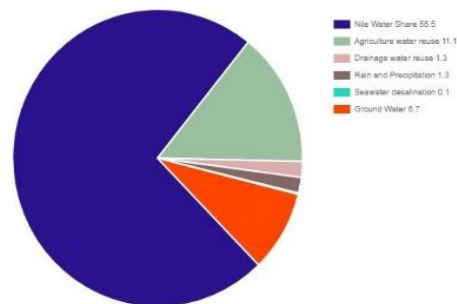


Chart 3

about 90% of its freshwater needs. However, increasing demand, pollution, and the potential impacts of climate change and upstream developments (like the Grand Ethiopian Renaissance Dam) pose significant challenges.

Water sources problems:

1. Dependence on the Nile River:

- a) Limited Alternative Sources
- b) Upstream Threats

2. Water Scarcity:

- a) Arid Climate
- b) High Demand

3. Pollution:

- a) Industrial and Agricultural Runoff
- b) Household Waste
- c) Salinization

4. Climate Change:

- a) Reduced Rainfall
- b) Extreme Weather Events

Solving challenges:

1. Invest in Alternative Water Sources
2. Invest in Alternative Water Sources
3. Combat Pollution
4. Adapt to Climate Change

Prior solutions already tried:

1. National Water Resources Plan (NWRP)

For Modernizing irrigation systems, Expanding wastewater treatment and reuse, and promoting water conservation practices.

2. Public Awareness Campaigns

The government has launched campaigns to educate citizens about the importance of saving water and adopting efficient practices, and programs to train farmers in water-efficient agricultural techniques have been implemented.

3. Wastewater Treatment and Reuse

Egypt has built numerous wastewater treatment plants to treat sewage and industrial effluents, making the water safe for reuse in agriculture and industry, and one of the largest wastewater treatment plants in the world, located in the Sinai Peninsula, treats agricultural drainage water for reuse.

Increasing industrial base In Egypt.

Although Egypt has strategic location, large workforce and growing economy, Egypt needs to make large and strong industrial base.

Industry is one of the important properties to classifying all countries, is it strong or not?

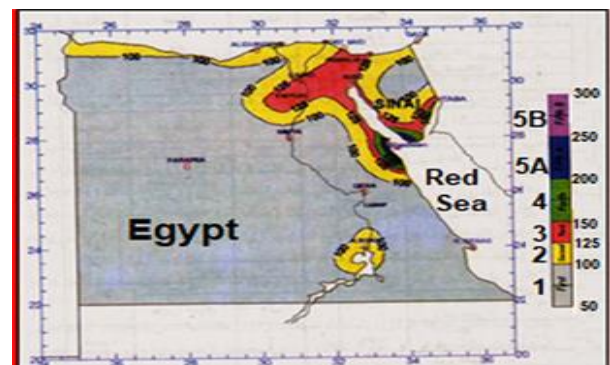
In Egypt, there is 147 industrial area, and its area is 22.6 million M^2 . For area of Egypt, this area is few, and it makes small ratio as industrial base, which it effects on the Egyptian economy. For making Egyptian economy stronger is increasing industrial base.

So, there are some solutions such as

- Exploitation of desert areas
- Infrastructure and supply chain development
- Workforce development and human capital

Exploitation of desert areas: Egypt has 94% of desert areas so, we need to exploit these areas for agriculture, tourism and industry. Some industries must be far away from urban cities such as energy, mining and construction, heavy and light industries, agriculture and eco-friendly industries.

Infrastructure and supply chain development:



Graph 6

Firstly, improving logistics networks (ports, railways) to lower costs and support local raw materials production to reduce dependence on imports.

Workforce development and human capital: promote vocational training for high-demand skills, encourage STEM education to support innovation and reform technical education to align with industry needs.

By attention to these ideas, Egypt will become country which has strong economy.



Figure 8

Increasing opportunities for Egyptians

Increasing work opportunities is important case for Egypt which Egyptians are main reason to provides Egypt at every field. So,

Figure 4

Egypt must interest at this problem. Now what do reasons which Egyptians stay for it? There is a strong economy, an attractive living environment and the most important is better job opportunities.

Strong economy: to achieve it a country should be interested in tourism, industry and agriculture and that is by Strengthening the job market. So, these are two suggestions for it.

1. **Expanding industrial and manufacturing sectors:** more factories and industries mean more jobs. The government should invest in local production to reduce reliance on Imports and create skilled employment.

2. **Enhancing tourism and hospitality:** developing new tourist attractions and improving services to generate more employment.

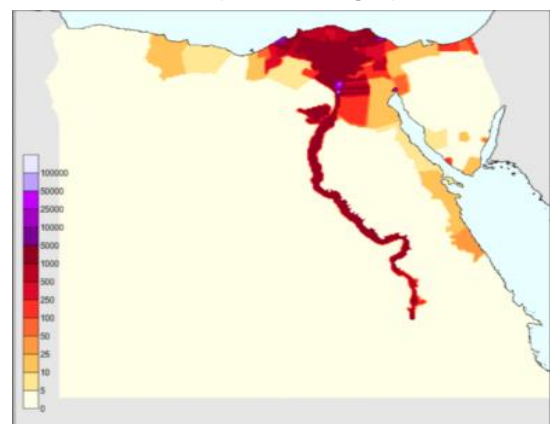
An attractive living environment: this factor may be main reason to Egyptians stay. So, when Egypt provides better living environment, Egyptians do not need to go out of Egypt. To providing better living environment for Egyptians should be there

1. **Improving public services:** invest in health care, education and housing to improve quality of life
2. **Ensuring safety and stability:** a secure environment encourages professionals to stay and invest in their future locally.

Better job opportunities: the main factor of this grand challenge. Before, two factors were about how makes Egyptians stay in Egypt. But this factor about how Egypt provides jobs for many people. For achieving it, firstly, Egypt should encourage foreign and improve special economic zones then, improving salaries by raising minimum wages, providing job security and supporting small and medium enterprises

The Arid Areas

arid areas is a huge problem that Egypt faces. As about 95% of the Egyptian population live on the river Nile as you see in graph 6.



Graph 7

Now, let's learn about the bad impact of this grand challenge as it has lots of problem like:

1. **Water Scarcity**
2. **Desertification & Land Degradation**
3. **Extreme Climate Conditions**
4. **Limited Economic Opportunities**
5. **Transportation & Accessibility Issues**
6. **Environmental Risks**
7. **Security & Border Challenges**

Of course, we need to focus on how to solve this problem so here is some keys ideas to solve this problem:

1. **Water Management & Conservation**
2. **Combating Desertification**
3. **Climate Adaptation Strategies**
4. **Economic Development & Job Creation**
5. **Improving Transportation & Infrastructure**

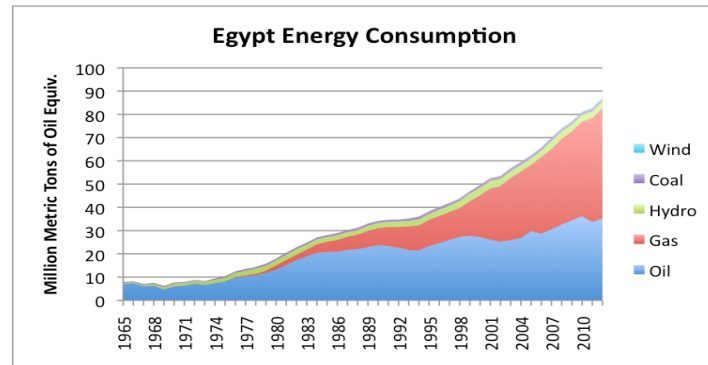
and now, here are two solutions that we have already tried:

1. **Environmental Protection**
2. **Enhancing Security & Border Protection**

Problem to be solved

Energy may be one of the most important materials in our new age. Now we have many energy sources, and many classifies for energy sources such as:

- 1) **non-renewable:** coal, natural gas and



Graph 8

petrol

- 2) **renewable:** solar energy, wind energy, nuclear energy and water energy.

Energy plays the main role in many fields and daily activities such as:

- 1) **Health care field:**

Which medical, equipment, emergency rooms and laboratories require continuous power, and used in vaccine storage and pharmaceuticals.

- 2) **Transportation field:**

All public transportation like (metro systems, electric buses, railway networks) depend on electricity and fuel.

- 3) **Economic field:**

- a) **Industrial sector:** factories rely on electricity and natural gas to operate machinery and production lines.
- b) **Agriculture sector:** Many activities depend on energy such as irrigation systems, food processing and cold storage and transportation.

- 4) **Public life:**

Energy is used in everything in our homes for lighting, air conditioning and appliances relief.

Uses Of Electricity In Our Daily Life

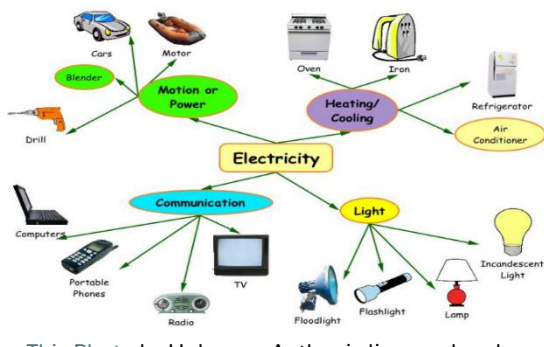


Figure 9

5)Technological field:

Energy is the main factor for the idea of technology, and the internet, mobile networks and data centers require uninterrupted power. Then, producing energy is the most important challenge we should be interested on it. There are many new ways to produce energy like

- Innovative vertical axis wind turbines
- Biofuel
- Microbiological cells
- Liquid fuel cells
- Micro sized hydro power station
- Small solar chimney
- Pumped hydro storage

Between these ways and after searching, we discovered the best way to produce energy, and it is innovative vertical axis wind turbines. This idea has many pros and properties such as

➤ Design and structural:

- a) can capture wind from any direction without requiring realignment.
- b) Suitable for urban environments, rooftops and small-scale applications

- c) Lower center of gravity provides better stability
- d) Lightweight and durable materials

➤ Efficiency and performance:

- a) better performance in low wind
- b) Innovative blade designs
- c) Self-starting capability
- d) Improved energy conversion

➤ Environmental benefits:

- a) Low noise levels
- b) Minimal impact on wildlife
- c) Sustainable energy solution

➤ Installation and maintenance:

- a) Easier installation
- b) Lower maintenance costs
- c) Modular scalability

After these properties, we chose the innovative vertical axis wind turbine to make our capstone

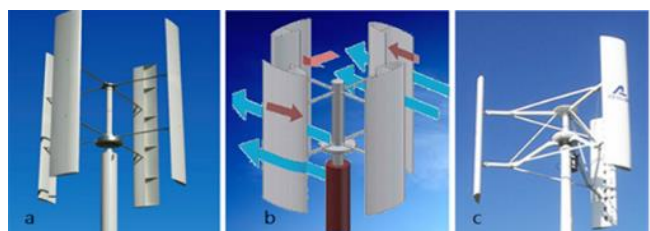


Figure 10

Research

Topics related to the problem:

1. renewable Energy Sources:

- Solar, wind, hydropower, geothermal, and biomass energy.

2. Energy Storage Technologies:

- Battery technologies (lithium-ion, solid-state, flow batteries).
- Hydrogen storage.
- Gravity-based energy storage.

3. Energy Harvesting:

- Piezoelectric systems.
- Thermoelectric energy.
- Kinetic energy recovery systems (KERS).

4. Decentralized Energy Systems:

- Microgrids.
- Community-based renewable energy projects.
- Off-grid solar and wind systems.

5. Alternative Fuels:

- Biofuels (ethanol, biodiesel, and algae-based fuels).
- Hydrogen as fuel.
- Synthetic fuels and e-fuels.

6. Marine Energy:

- Tidal energy.
- Wave energy.
- Ocean thermal energy conversion (OTEC).

7. Smart Energy Management:

- Smart grids and IoT for energy efficiency.
- Demand response technologies.

Topics related to the solution:

1. Hybrid Renewable Energy Systems (HRES):

- Combines VAWTs with solar, hydro, or batteries for continuous power.
- Ideal for homes, businesses, and remote areas.

2. Off-Grid & Urban Energy Solutions:

- *Uses VAWTs in microgrids for villages, rooftops, and street lighting.*
- *Reduces dependence on fossil fuels in urban and rural areas.*

3. Energy Storage & Smart Grids:

- Stores excess wind energy in batteries or hydrogen fuel cells.
- Smart grids optimize energy distribution for stability.

4. Transportation & Charging Stations:

- VAWTs provide clean energy for EV charging stations.
- It can be placed on highways, ports, and railway hubs.

5. Sustainable Agriculture & Rural Electrification:

- Powers irrigation, lighting, and small industries in farms.
- Replace expensive diesel generators in rural areas.

6. Floating & Offshore Wind Farms:

- Uses VAWTs on lakes and oceans for renewable offshore energy.
- Solves land-use issues and harnesses stronger coastal winds.

Other solutions already tried

Previous tried solutions in the World and Egypt:

1. Solar Power Projects :

Due to its location along the global solar belt, Egypt has adopted solar energy as a strategic objective. The country has remarkably large solar energy projects, such as the commissioning of a 140 MW solar thermal plant in Kuraymat (2010). In addition, Egypt has explored both Photovoltaic (PV) and Concentrated Solar Power (CSP) technologies. PV may be the better choice, but CSP is likely cheaper and a more natural option for Egypt's hot climate. Small-scale economies have generated local jobs and off-grid solutions solar projects, such as those in rural areas.

2. Biogas Technology for Rural Development

The Egyptian government has adopted biogas technology for rural development. The second, the Bioenergy for Sustainable Rural Development Project, aims to produce biogas from animal and agricultural waste, which in turn replaces butane gas for cooking and heating. This initiative has reached more than 5,000 rural people, established 18 family-sized biogas units, and trained 40 individuals in biogas technology. Biogas is a renewable energy source, and also helps to eliminate air pollution from burning crop residues.

3. Hybrid Energy Systems

The integrated systems combining multi-renewable energy sources (solar, wind, and biomass, etc.) with energy storage systems (ESS) are developed. So you are able to store extra energy when renewables generate a lot and release it when renewables generate little. For instance, the hybrid systems deployed in Egypt and Tunisia are enhancing grid stability and enabling remote area energy access.

4. Large-Scale Solar and Wind Projects

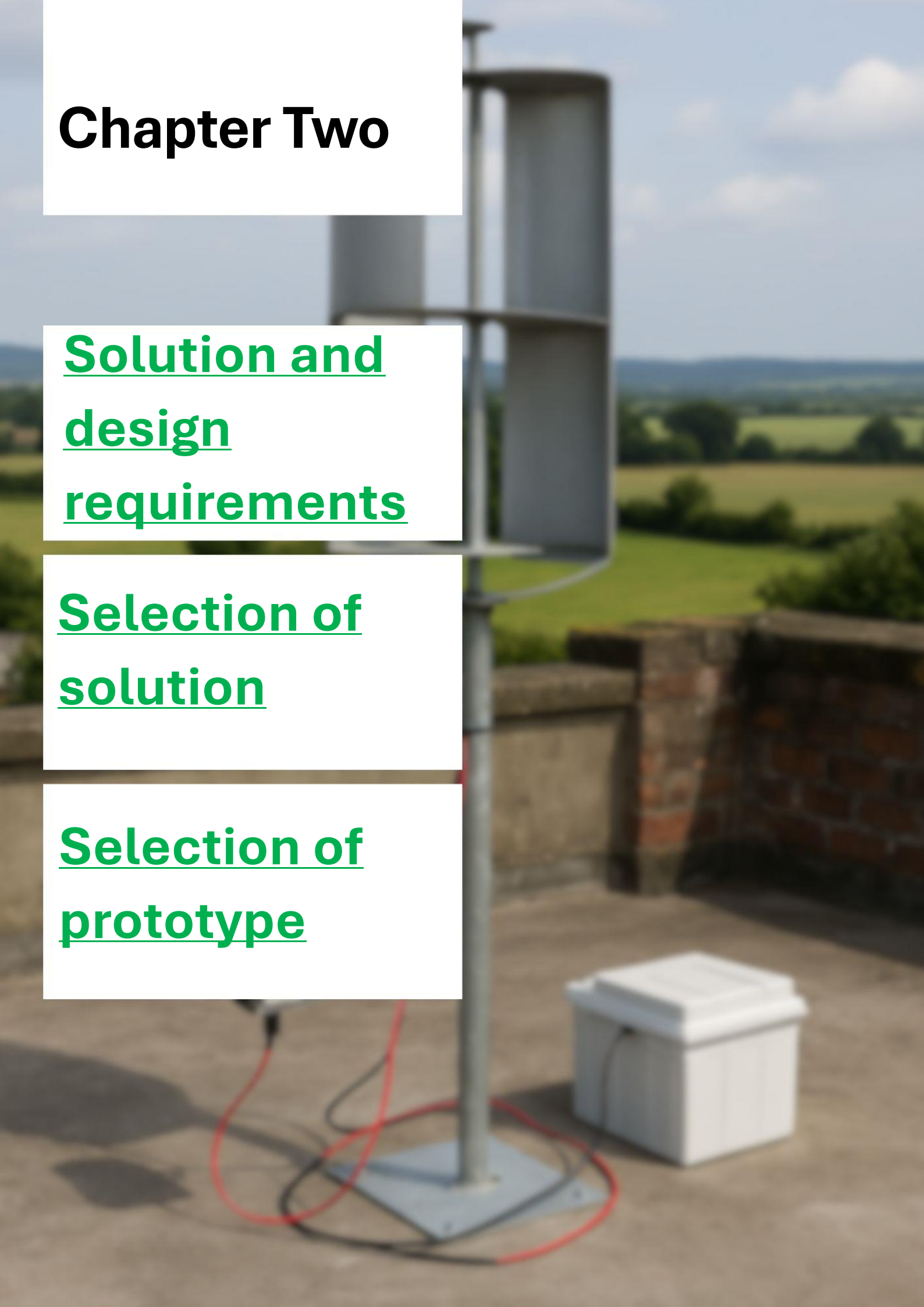
Countries have also invested heavily in large-scale solar and wind farms to power everything from homes and businesses to electric vehicles. Germany, for example, added 14 gigawatts (GW) of solar capacity in 2023, a record and 56% above target. Solar power generation in the U.S. is projected to increase by 41% in 2024, aided by federal and state support schemes. Moreover, and the Red Sea through the commission of large solar projects, including the 140 MW solar thermal power plant as well as wind farms along the Red sea coast.

Chapter Two

Solution and
design
requirements

Selection of
solution

Selection of
prototype



Solution and design requirements

All successful projects need some suitable requirements as criteria of success. Some requirements were used for making the prototype such as prototype's dimensions, amount of generated power and materials used. In our capstone this semester, it's about producing electricity and storing it. For making our prototype more useful for our environment we tried to not produce any pollution. So, materials that is used should be highly eco-friendly that's reducing pollution and protecting our environment.

As a result of that, the selection of the way to produce energy was implementing a vertical axis wind turbine.

Firstly, as was mentioned in the challenge, the prototype's whole volume shouldn't exceed 0.5 m^3 , to achieve that design requirement the vertical axis wind turbine was put in a wooden frame of $0.75 \text{ meter} \times 0.75 \text{ meter} \times 0.5 \text{ meter}$ dimensions to restrain the prototype's volume, knowing that there are much space in the frame so the prototype is 100% doesn't exceed 0.5 m^3 .

Secondly, to verify that the prototype produces power of 40 watt-seconds in at most 5 minutes, we ensured the production of at least 0.133 watts in each second along the 5 minutes, according to the calculation of minimum power as follow:

$$\text{Power} = \frac{\text{Energy}}{\text{time}} = \frac{40 \text{ watt-seconds}}{5 \text{ minutes} \times 60} = 0.1333 \text{ watt}$$

The vertical axis wind turbine generated much more with generating average energy of 105.54 Joules (watt-seconds) in 5 minutes, and producing the needed 40 Joules in 1.9 minutes.

Thirdly, for making the prototype the selection of environmentally friendly materials was a priority, so the blades were cut vertically from a PVC pipe which is made of **vinyl chloride monomers** ($\text{C}_2\text{H}_3\text{Cl}$), a highly environment friendly material. Also, the frame and the shaft was made wooden, and finally, the batteries used was chosen to be lithium-ion batteries. All these materials used are highly eco-friendly.

Selection of solution

Selection of solution for solving the challenge is crucial, so the choose of solution between the other solutions came after a deep research in every possible solution, and on specific criteria as:

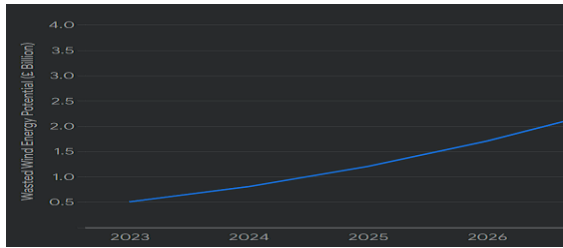
- 1) Availability of the energy type which will be exploited
- 2) The project's overall costs
- 3) Using environmentally friendly materials
- 4) Effectiveness of the project
- 5) Specifying the design requirement.

● Availability of the energy type which will be exploited

From very different types of clean energy as solar, hydropower, wind, and biomass, wind was the best choice to be easily implemented in Egypt, since Egypt as a country has powerful resources of wind energy.

Production of electricity from wind is a thing that not all know its great benefits, a lot of wind energy is daily not given enough attention and is wasted, from there came our solution which is building and implementing an innovative

vertical axis wind turbine. That project can make a revolution in Egypt's energy sector.



Graph 9 shows the wasted potential of energy production in Egypt

● The project overall costs:

Another thing to involve in the selection of our solution, after searching and comparing the other solutions we found that the innovative wind turbine may be the lowest in costs to be implemented specially in Egypt, so buying the used materials ensuring the lowest cost possible was crucial, as a result of that the project was implemented with the lowest implementing costs possible.

● Environmentally friendly materials:

Ensuring that the project is fully eco-friendly or release the lowest possible greenhouse gases and offering an alternative of fossil fuels was an important step of our plan in applying this project, so finding alternatives from the environment or finding recyclable materials was from our priorities.

● Effectiveness of the project:

After a deep search into types of vertical axis wind turbines (VAWT), we get to know that there is two types of VAWT which are **1) Savonius Wind Turbines:** which is suitable for lower wind speeds, and lower efficiency of energy production, **2) Darrieus Wind Turbines:** this type is more suitable for higher wind speeds, as a result of that it is more efficient in electricity production.

The best method to achieve the highest effectiveness is to implement what is more suitable for implementation in Egypt which is the Savonius type.

● Specifying the design requirements:

The last criteria is to easily achieve and specify the desired design requirements, in all the proposed solutions, implementing a VAWT is perhaps the most suitable solution for meeting the design requirements.

In conclusion, our project offers an effective, practical, low-cost, and hands available solution for the energy crises that faces Egypt and draining its economy.

Selection of prototype

Selection of prototype was not an easy task. As we needed to choose carefully everything of our prototype and need to solve a lot of problems that faces us while building the prototype.

We need to build a prototype that can generate 0.04Kws in at most five minutes, so we decided to work on a prototype of a savonius vertical wind axis turbine. Savonius wind turbines has a lot of advantages. As this type of wind turbines gives good efficiency and don't need a strong wind to start rotate and generate electricity.

As I mentioned our prototype is a savonius wind turbine that built on small scale. So we build our prototype from other materials that is not used in the big scale ones. We decide to build our prototype from eco-friendly materials.

- 1. The DC motor:** we used a DC motor that to generate electricity. The DC motor is 12-24 volt and 58000 RBM. We chose a heigh voltage motor to make it easier to generate electricity with less effort and we used low RBM because our prototype doesn't need heigh RBM

which make our prototype cost less money.

2. **The blades:** We build it with PVC pipes Because it is cheap and light which make this part work with good efficiency.
3. **The axe:** we made the axe from wood as wood is a cheap light material that will help the prototype rotate faster. We made its hight 60cm and its connected to two ball bearings
4. **Ball bearing:** ball Bering function is to connect the shaft of the wind turbine with the frame and it work as a holder for the axis where the axis can rotate freely.
5. **Frame:** we made a frame to out prototype that is made with wood. The frame function is to give stability to our prototype.
6. **The batteries:** we used four lithium ion batteries. The voltage of each one are 3.7V so it can handle the voltage that the DC motor makes.

Egypt needs a lot energy to meet the need of the population.

How the prototype meets

Design requirements

To make any prototype successful it should be built on specific design requirements that make sure that the prototype offers its function. So we built our prototype on specific design requirements such as:

1. **Avoid using hydrogen:** we decided to build a prototype that doesn't need a hydrogen to work as hydrogen is not a safe matter to deal with.
2. **The maximum volume is 0.5m³:** we decided to make our prototype volume 0.5m³ because if we wanted to build in a bigger scale, we could build it in places that have wind and do not have big spaces to build it there like Cairo.
3. **Fast in generating electricity:** we build our prototype in a way that make it can offer more electricity in a small time. As

Chapter Three

Methods and materials

Test plan

Data collection



Methods and materials

Vertical Axis Wind Turbines (VAWTs) are wind energy systems in which the rotor spins around a vertical axis, allowing them to capture wind from any direction. And the Engineering design process (EDP) was the steps that we move on to construct our prototype.

Key principles:

1. The research:

- We do a lot of research to collect large amounts of information about our prototype and how to construct it.

2. Construction of a frame:

- The frame is 0.75 meters wide, 0.5 meters high, and 0.75 meters long. The frame was made from wood.
- The frame was constructed by iron nails with a hammer.

3. Design the Savonius rotor:

- We cut a PVC pipe into two halves "half-cylindrical shape" and constructed it on a vertical shaft the vertical shaft has a length of 54 cm. So, the blades take the "S" shape.
- We have punctured a hole in the vertical shaft at the bottom, to connect the DC motor to the vertical shaft through a recycled hand blender coupler.
- The PVC pipe has a length of 30 cm and a width of 10 cm, after the construction action, the blades have a 20 cm diameter (every half of the two halves is 10 cm diameter).

4. Add the Ball Bearing:

- We constructed two Ball bearings (at the top and the bottom). The smaller one is 6202 and the bigger one is 6302. The ball bearings are placed opposite each other.

5. Connect the DC motor to the batteries:

- We connected a Schottky Barrier Diode to 12V-24V and 5800 RS550 DC motor to prevent the current reverse of the electricity to the DC motor and We connected the battery to the Diode in the DC motor and started to store the electricity.

6. Final touches and testing:

- We tested the prototype many times to make sure it produces the required energy needed.
- The prototype generates 105 joules as an average production.
- We checked the mechanism of the prototype for smooth movement.

Materials



Name: Wood

Quantity: 15 pieces

Cost: 530 L.E.

Usage: Build the frame and the vertical shaft.



Name: Recycled Li-ion battery

Quantity: 1 piece

Cost: 30 L.E.

Usage: Energy storage



Name: Ball bearings

Quantity: 2 pieces

Cost: 95 L.E.

Usage: Smoothing the rotation of the vertical shaft.



Name: Battery Holder

Quantity: 1 piece

Cost: 8 L.E.

Usage: Holding the battery



Name: PVC pipe

Quantity: 1 piece

Cost: 170 L.E.

Usage: Building the blades



Name: Diode 1N5819 TH

Quantity: 1 piece

Cost: 1 L.E.

Usage: Prevent the reverse current



Name: DC motor

Quantity: 1 piece

Cost: 250 L.E.

Usage: Act as a generator



Name: Recycled Hand-Blender's coupler

Quantity: 1 piece

Cost: 20 L.E.

Usage: connecting the vertical shaft to the DC motor



Name: Terminals

Quantity: 2 pieces

Cost: 5 L.E.

Usage: connecting wires to the battery

Test plan

As a part of any professional project, the project must be tested by accurate tools and unique design requirements to determine the efficiency of the project, This Capstone project has specific requirements, which are:

1-Speed of generating energy, as the project must produce 0.04 kW in at most 5 minutes

2-Storing the energy in any storage device to be used anytime.

First trial:

We added a ball bearing in the first test. After testing the prototype with one ball bearing, we discovered that the blades vibrate strongly. The shaft is short, so we decided to change the shaft to a longer one and add a second ball bearing to reduce the vibration and give stability to the blades and the prototype. So, the prototype is not able to generate the required energy.

Second trial:

After additional materials were in the 1st trial, the prototype was tested again, and this time, there was a problem. The problem was human error, as the air source used was the air blower. After the required time, the project didn't achieve the required energy needed due to measurement errors. So, we tried to reduce human and measurement errors.

Third trial:

After the two past trials, our prototype is supposed to achieve the required energy. We measured the energy generated by the multimeter. So, the prototype generated 105.5 joules as an average production and connected the motor to a 3.7V battery.

After these three trials, we noticed that the prototype was qualified enough to achieve the design requirements, as the prototype produced 105.5 joules in 5 minutes, the prototype produced 40 joules in a time of 1.9 minutes, and these were the concluded data:

Trial number	Time	Current intensity	Potential difference	Power	Energy
First trial	5 minutes	0	0	0	0
Second trial	5 minutes	0.1152±0.0022 Ampere	1.0742±0.0137 Volts	0.1237±0.0039 Watts	37.110±1.17 Joules
Third trial	5 minutes	0.274±0.0046 Ampere	1.292±0.016 Volts	0.354±0.010 Watts	105±3.0 Joules

Table 1

Data collection

After conducting the test plans, here are the outcomes:

For the first test plan:

After finishing constructing the prototype, the prototype was tested, and we could not measure the potential difference and the current intensity, because the shaft vibrates too much, and did not rotate correctly. So the concluded modifications were that there was an additional ball bearing must be added to reduce the turbine vibration. And also, we must make a longer shaft, because this shaft was too short.

For the second test plan:

The prototype was tested again after doing the modifications, so the energy was measured using the multimeter, but there was a problem that faced the prototype, which was human error, so the average measurements were:

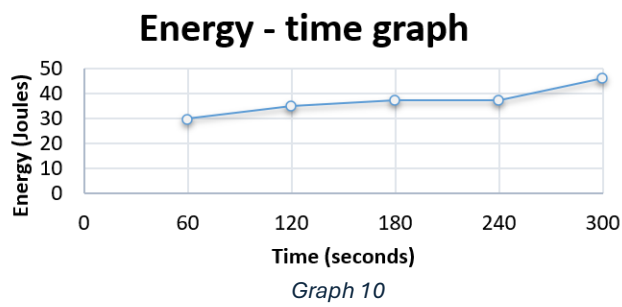
Time	Current intensity	Potential difference	power	Energy
5 minutes	0.1152±0.0022 Ampere	1.0742±0.0137 Volts	0.1237±0.0039 Watts	37.110±1.17 Joules

Table 2

For the third test plan:

After the two past trials, the prototype is supposed to achieve the required energy. We measured the energy generated by the multimeter. So, the prototype generated

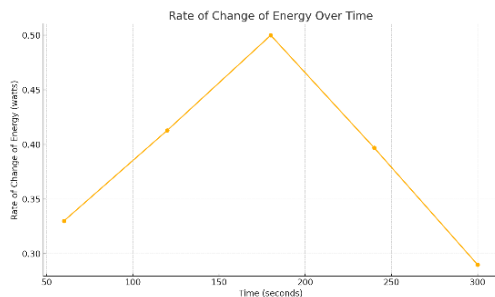
105.5±3.36 joules as an average production, and also we graphed the data:



From the noticed and the collected data in the project, we noticed that there is a direct relationship between time and energy. This is the data table for the rate of change of power over time:

Time (in seconds)	Rate of change of power
60	0.330 watt
120	0.413 watt
180	0.500 watt
240	0.397 watt
300	0.290 watt

Table 3



Graph 11

Here are the used tools:

Tool	Usage	Quantity
Multimeter	Measuring the generated energy	One piece
Hammer	Constructing the wooden frame	One piece
Wood punch	To perforate the shaft for the blades	One piece
Air blower	To test the prototype	One piece
saw	To cut the PVC pipes into small blades and to cut the wood for the frame	One piece

Table 4

Chapter Four

Analysis and
discussion

Recommendations

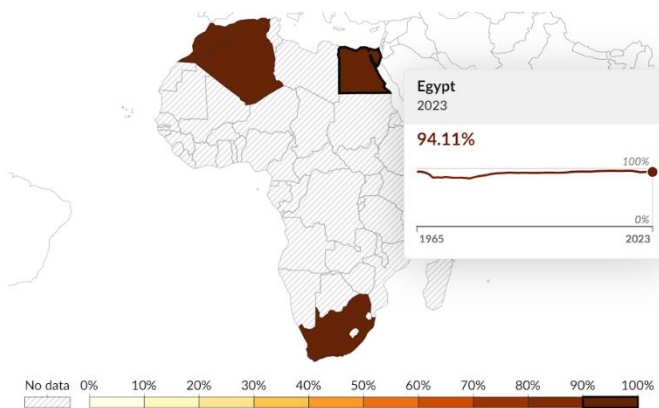
Learning
outcomes

Literature
cited

Analysis and discussion

Introduction:

Egypt had been confronting several energy crises. This capstone project solves these crises practically. Challenges and crises like these are related to the SDGS, Egypt's 2030 vision, and the 11 grand challenges of Egypt. Crises related to the SDGS include Providing Affordable and Clean Energy (SDG 7), Responsible Consumption and Production (SDG 12), and Climate Action (SDG 13). The solution this project offers also contributes to the shift to renewables as one of the strategies of Egypt's 2030 vision, and to the 11 grand challenge of Egypt, such as: Improving the scientific and technological environment for all, addressing & reducing pollution in air, water and soil, reducing and adapting to the effects of climate change, and the need to improve the use of alternative energies. The last challenge is the focus challenge of Egypt's development economically, socially, and politically. In the year 2023, Egypt's share of primary energy consumption from fossil fuels was about 94.11%, and only 5.89% was the share of primary energy consumption from renewable sources, as shown in the chart (10).



Graph 12

So, as in this capstone project, implementing a vertical axis wind turbine (VAWT) offers the solution for Egypt's energy crisis by capturing wind energy through blades that moves with rotational motion, which is transferred to a DC motor that works as a generator, and lastly stores the energy in a lithium-ion battery. Also, this system is privileged with a frame for stabilization against vibrations and rigid support.

Calculations, Laws, and Rules:

As this project generated an average of energy produced of 105.5 Joules, a process was gone through to calculate this based on scientific laws and rules, while taking needed cautions and taking measurements in laboratories using the suitable tools.

1st: Calculation of the velocity of the wind influencing the blades:

Bosch GPL 620 was used as a wind source, the wind speed influencing the blades was need to be calculated to ensure the practicality of the project. To calculate this, the Jet Velocity Decay Formula was used as follows:

$$V(x) = V_o \times \left(\frac{D}{x}\right)^n$$

Where:

- $V(x)$: Velocity at distance X.
- V_o : Initial Velocity, value of 75 m/s.
- n : Exponent depending on jet type, value of 0.57 for Bosch GPL 620.
- x : Distance from the nozzle, value of 30 cm.

$$V(x) = 75 \times (0.03 / 0.3)^{0.57} \approx 20 \text{ m/s}$$

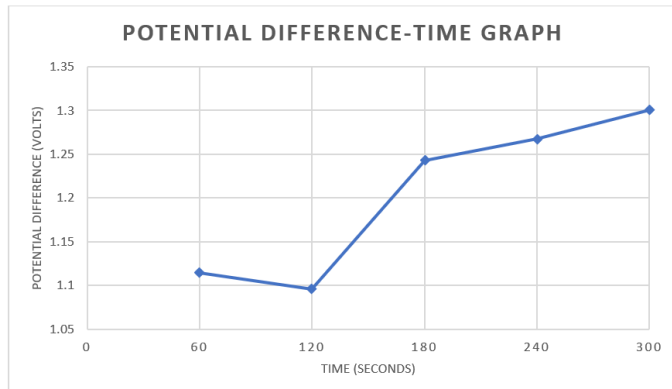
With a wind speed like this, the prototype can easily generate more than the produced in higher wind speeds, as in the western desert, which has better wind conditions for implementing this project with wind speeds ranging from 27 m/s to 33 m/s.

2nd: Calculation of Potential Difference:

A measurement was taken every 1 minute for 5 minutes using a multimeter.

Potential Difference (Volts)	1.134±0.014	1.186±0.015	1.294±0.016	1.570±0.019	1.275±0.016
Time (seconds)	60	120	180	240	300

Table 5



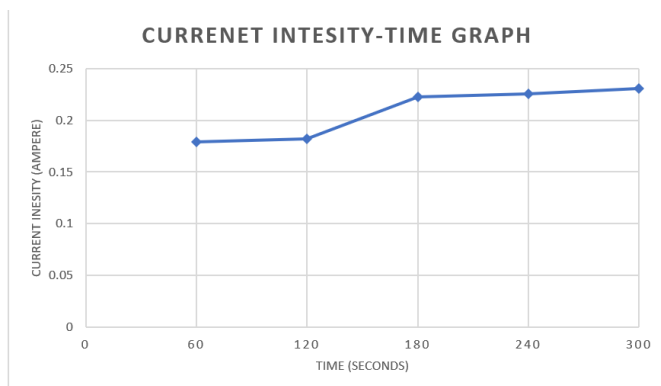
Graph 13 shows the Potential difference as a function of time over a time interval of 5 minutes

3rd: Calculation of the Current Intensity:

A measurement was taken every 1 minute for 5 minutes using a multimeter.

Current Intensity (Ampere)	0.258±0.004	0.266±0.004	0.288±0.004	0.258±0.004	0.300±0.005
Time (seconds)	60	120	180	240	300

Table 6



Graph 14 shows the Current intensity as a function of time over a time interval of 5 minutes.

4th: Calculation of Power:

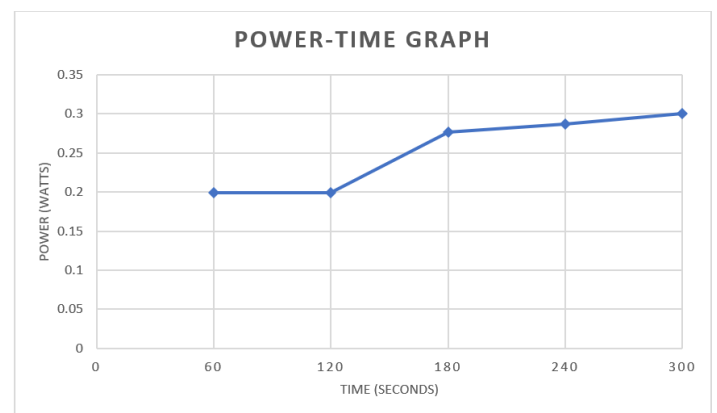
This was calculated by using the Electrical power formula. $P = I \times V$

Power (watts)	1.134x 0.258	1.186x 0.266	1.294x 0.288	1.570x 0.258	1.275x 0.300
Time (seconds)	60	120	180	240	300

Table 7

Power (watts)	0.290±0.006	0.310±0.006	0.372±0.007	0.405±0.008	0.382±0.008
Time (seconds)	60	120	180	240	300

Table 8



Graph 15 shows the Power as a function of time over a time interval of 5 minutes.

Average Power (watts)=

$$\frac{0.290+0.310+0.372+0.405+0.382}{5} = 0.3518 \pm 0.0074$$

watts

5th: Calculation of the average generated energy: To calculate the average generated energy, **Joule's Law of Electrical Energy** was used. $E = P \times t$

used. $E = P \times t$

• $E = 0.3518 \times 300 = 105.54 \pm 2.22 \text{ Joules}$

6th: Calculation of the battery capacity: To calculate the battery capacity to ensure its ability to store the generated energy in 5 minutes by using a 3.7v, 500mAh lithium-ion battery, the following steps were taken:

1) Calculate Energy (E) in Watt-hours (Wh):

$$E = V \times Q$$

$$E = 3.7 \times 500 \times 10^{-3} = 1.85 \text{ Wh}$$

2) Convert to Joules:

From the relation $1 \text{ Wh} = 3600 \text{ J}$

$$E = 1.85 \text{ Wh} \times \frac{3600}{1} = 6,660 \text{ Joules}$$

. After the previous calculations, we ensured that the battery could store 40 Joules and more by far.

increasing the wind speed which will increase the prototype efficiency.

5. **Research on better materials:** for further research, we recommend make a research on cheap strong materials that will lower the cost of our project and increase it's efficiency.
6. **Make a smaller frame:** we recommend making a smaller frame because this will low the cost of the cost of the prototype.

Recommendations

As everything in our world changes and, science evolves day after day. Here are some recommendations that will modify our prototype.

1. **Gear system:** we recommend using a gear system that connect the DC motor with the shaft of the wind turbine. As gear system will work on increasing the number of the RPMs of the DC motor, which will increase the efficiency of the prototype.
2. **Bigger blades:** we recommend using bigger blades than the ones that we used. As there is a direct relation between the blades size and the RPMs that the prototype makes.
3. **Replacing the building materials in the large scale:** we recommend using different materials while building the larger scale. For example, we can make the blades from sheet metal instead of PVC pipes. We can make the shaft of the wind turbine from aluminium or carbon Fiber instead of wood.
4. **Making a system to increase the wind speed:** we recommend making a pipe shape with a large side faces the wind and a small side that faces the wind turbine. This shape will work on

Learning Outcomes

Subject	Usage
PH.1.08	Helping us at knowing properties that affect air flow as atomic pressure, temperature and others.
PH.1.09	Helping us with calculating energy produced and design blades by calculating the amount of air (Pascal's Principle) and knowing air flow properties or factors.
ME.1.06	Helping us with design blades, select angles between it and its number (Lami's law) for making equilibrium between forces.
ME.1.09	Helping us at calculating force (centripetal force) affects blades during its movement.
ES.1.08	Helping us at selecting materials are used at prototype and knowing properties of ores that can help us.
ES.1.09	Classifying energy sources as renewable and non-renewable and knowing many ways how generate energy from all energy sources.
ES.1.10	Helping us by explaining a lot of details about producing electricity from wind, and all ways for generating energy needed.
MA.1.07	Helping us at making graphs and relations for results by using parabola and quadratic functions.
CH.1.13	Helping us at understanding and studying electricity deeply and its relationship with chemistry.
CH.1.14	Storage electricity at batteries and select a type of battery (galvanic cell or electrolytic cell) and what are pros and cons for every cell.
BI.1.13	Helping us at realizing importanceenergy source's type for producing electricity and its effects on environment.

Table 9

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